



A system dynamics gamification tool to enhance supply chain resilience in times of disruptions

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Abstract

In recent years, significant disruptive events have stressed the global business environment and supply chains worldwide, and these events have become more disturbing and frequent. Consequently, organisations have been facing many troubles in different echelons of their supply chains, such as high inventory stockouts or unplanned supply and recovery costs, significantly affecting their performance and stability. Accordingly, strategists must acknowledge enterprise and supply chain resilience as a vital competence in organisational strategies and ploys to guide decision-makers when dealing with disruptive events. This work aims to help organisations develop a powerful approach for testing and improving a decision-making process to minimise stockouts, through a simulation game that considers essential supply chain components, such as suppliers, products, and disruptive events, coexisting in a dynamic system. Upon implementing the proposed game within an organisation, the main expected outcomes include: (i) enhanced decision-making capabilities; (ii) increased awareness and preparedness to address disruptive events; (iii) improved practical understanding of resilient supply chain management practices; (iv) cost-effective training and awareness-building; and (v) repetitive learning opportunities, thereby enhancing supply chain management and resilience. The game's simplicity is crucial to ensure that this tool remains accessible and practical for its intended audience, as excessive complexity may potentially discourage its use.

Keywords Gamification · Enterprise resilience · Supply chain resilience · System dynamics · Simulation

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1 Introduction

Classical strategic approaches like cost reduction and efficiency must now coexist with Enterprise and Supply Chain Resilience (ESCR) in a balanced organisational strategy, and this has become evident since disruptive events (DEs) have been growing throughout the business and logistics world. A few years ago, we lived through a dangerous pandemic that, besides the severe health problems, generated many negative consequences in supply chains worldwide, such as extended lead times, increased logistics costs, and lower inventory availability. The pandemic had not ended, and a war exploded between Russia and Ukraine, adding uncertainty and costs to critical raw materials for many companies. While this war is still going, another war burst in the Middle East, expanding risks, and further complicating global supply chains. The mentioned DEs are very notorious and threatening. Still, organisations also frequently face other events that must be addressed, such as unexpected demand increases or falls, changes in macroeconomic factors affecting costs, and suppliers' backorders. Both types of events require appropriate training because the workforce needs to know the course of action in the presence of DEs. Enterprise Resilience (ER) is the capability to anticipate and be prepared to face disruptive events and, if their occurrence is inevitable, the capacity to recover as soon and efficiently as possible (Sanchis and Poler 2014). Lack of awareness and knowledge in resilience theory among individuals at any stage of an SC can lead to deeper troubles, such as increasing stockouts and reducing profits. A cost-effective way to train individuals and to create awareness about resilience, is through gamification, since it can model uncertain real-life situations within a game environment, without risking real profits and service levels. Gamification is defined as the use of game-like elements in non-gaming contexts (Campos et al. 2024), it is a term derived from games, which includes activities with non-mandatory spatial and temporal participation, separate from the real world. Gamification has been among organisations for a long time, becoming a relevant mechanism to measure, develop, and enhance people's skills regarding strategic and tactical decision-making. Games are relatively inexpensive and easy to deploy since they do not require risking companies' assets by testing learners in real situations; instead, they can often work through simulations modelling existing situations. Gamification has been implemented in several fields, such as organisational strategy, supply chain (SC) management, logistics, information security, demand planning, recruitment, customer service, and tactical decision-making in general. It has been proven to be a valuable mechanism for people to learn and test their skills in different scenarios. One of its advantages is repetitiveness, and players can play as many times as needed to learn and get trained. Although this work aims to enhance resilience in SC management, it is important to refer to experiences in other fields, such as information security and manufacturing, since those fields can be sources of DEs. Filipczuk et al. (2019) explain how gamification can strengthen learning to avoid fraudulent phishing and to increase data security awareness. Additionally, Cole et al. (2019) recommend training to prevent data breaches suitable for any type of devices, and they state that human error is the main cause of these breaches. Regarding manufacturing, Paravizo et al. (2018)

explain that gamification mechanics can improve training and manufacturing education, even considering important advances such as innovation and Industry 4.0.

To implement gamification strategies, it is recommended to follow formal frameworks, Campos et al. (2024) propose a Canvas framework with six steps to gamify a process, and Wramsby (2016) analyses several frameworks, such as RAMP (relatedness, autonomy, mastery and purpose), MDE (mechanics, dynamics and emotions), and DMC Pyramid (dynamics, mechanics and components), and presents a framework on how to create a gamified experience, with seven steps.

Systems dynamics is a field known for its capacity to simulate real-life situations efficiently, and this is why an ESCR systems dynamics game, inspired by the classic Beer Distribution Game, is proposed in this work to test and improve people skills in a simulated SC where several DEs occur. Moreover, the main objective of this paper is to present an SC simulation game designed considering the underlying structural causes of system behaviour using system dynamics. This game was designed based on relevant research about ESCR.

Upon implementation of the proposed game within an organisation, the main expected outcomes are: (i) the enhancement in decision-making capabilities; (ii) heightened awareness and preparedness to face DEs in the SC; (iii) increased practical understanding of resilient SC management practices; (iv) cost-effective training and awareness-building that minimise risks to real profits and service levels; and (v) repetitive learning opportunities to reinforce key concepts and skills over time.

Thus, the game tests the management skills of individuals without incurring risk and at minimal cost. Moreover, the inclusion of system dynamics in the game allows players to use it repeatedly when necessary. The rest of this work is organised as follows: Sect. 2 is the literature review, which shows more details about the relation between gamification, system dynamics and ESCR, in Sect. 3 there is a detailed explanation of the game, including all the system dynamics formulations. The last section is Sect. 4, which contains conclusions and future research lines.

2 Literature review

Now, more than ever, it is clear that there is a large diversity of DEs, as mentioned by Sanchis and Poler (2014), moreover, authors in Sanchis et al. (2020) define DEs as foreseeable or unforeseeable events that directly affect an enterprise's usual operation and stability. Given the impact of all types of DEs on business stability and SC fragility, organisations need people with strong knowledge of ESCR through all organisational levels. The workforce first needs to understand and learn the main concepts regarding resilience in organisations to acknowledge DEs' risks, impact, and consequences; they need to be aware of the possible presence of those events and know what to do when they occur, depending on their degree of uncertainty; this implies that people must develop high decision-making skills towards creating absorption, adaptation, and regeneration capacities. Building an ESCR strategy and culture might lead to competitive advantages; this implies several infrequent investments; from a proactive perspective, organisations have to select which preventive

actions to invest in to anticipate very uncertain DEs. Preparedness is one of the primary constituent capacities of resilience.

It is important to know that several research works state that it is possible to estimate the ER potential; in Ewertowski et al. (2023), it is shown how to optimise this estimation based on machine learning methods. When organisations know their ER potential, they can focus on implementing proactive solutions to minimise the impact of potential DEs.

As stated before, organisations should invest in preparedness and anticipation capabilities. Still, even if they do, reality shows that predicting DEs remains a complex task. Thus, to overcome this situation, organisations must enhance the decision-making capabilities of their decision-makers. Training workers in ESCR with live DEs occurring in real situations would be hazardous and expensive since the related decisions can have serious consequences for the organisation. It is a fact that regarding DEs, real-time training has little room for repetitiveness because these events are infrequent and generally hard to predict. A potential solution comes with gamification, as it is accessible, risk-free, and cost-effective, and it can simulate some DEs to learn how to make the right decisions whenever they occur.

Experience using gamification to evaluate and train people is vast (Angles et al. 2021) since, for many years, it has been successfully tested in many fields, such as strategy creation, engineering, SC management, manufacturing, information technology (IT), health, recruiting, new product development, marketing, and demand planning. In Dick and Akbulut (2020), the authors analyse if Enterprise Resource Planning (ERP) simulation games can be applied to other management fields, concluding that this type of game guides decision-making with realistic practices. Another example is presented in Marcao et al. (2019), where the authors propose adopting gamification to improve quality, effectiveness, and efficiency. Authors in Gjertsen et al. (2017) also find that gamification can improve motivation and learning outcomes in security awareness and training programs.

The learning through games technique has also been applied to shape customers' behaviour; in Sterman et al. (2015), they present a system that motivates customers to reduce energy consumption through competition and collaboration using games. Sustainability has also been addressed through gamification; Mancebo et al. (2017) promote the creation of efficient and sustainable business processes, a green Business Process Management (BPM).

The concept of flight simulators has been widely used for teaching and training SC concepts and dynamics, along with other fields; Sterman (Sterman 2014) recommends using these simulators to teach essential ideas in areas such as business, strategy, and sustainability. These simulators enable players to learn experientially, exploring the consequences of different decisions so they can learn from themselves about the complex dynamics. These flight simulators for managerial situations have been successful because they can turn events into procedures when repeatedly trained (Sterman 2000). One of the best-known examples is the Beer Game from MIT Sloan School, which provides an opportunity for users to play the role of managers in a complex dynamic system (Sterman 2023). Like many other games, the Beer Game is developed based on system dynamics, a methodology to understand the change, utilising differential equations that allow studying the dynamics of a

system's set of available states. System dynamics aims to understand the structural causes of a system's behaviour (García 2023).

Formal training programs are necessary for organisations in the current complex business environment. As stated before, one recommended topic to include is ESCR, where gamification can play an essential role in adding value by implementing specialised simulation games.

After playing the proposed simulation game, players should question and evaluate their organisational strategy related to inventory management and investments in resilience, examining if efficiency and resilience are included as main components, aiming to create synergies; while efficiency is essential to ensure short and mid-term financial sustainability and profitability, resilience covers long-term viability (Arquer et al. 2022). It is advisable to incorporate strategies such as diversification of suppliers that can enhance resilience by reducing dependency on a single source; agile inventory management systems that enable efficient responses to fluctuating demand and supply disruptions, and contingency plans that provide a framework for swiftly adapting to unforeseen events, thereby bolstering the organisation's overall resilience and operational efficiency.

It is imperative to be aware of the fact that efficiency and resilience are not independent variables in an SC system; actually, they are highly interrelated, but there could be a trade-off between them; Arquer et al. (2022) evidenced the tensions between these two variables in a hybrid manufacturing/remanufacturing system, which uses a proportional order-up-to (POUT) replenishment policy, when exposed to demand shocks. Findings state that tuning the POUT policy to optimise efficiency may be problematic to the system's resilience.

The proposed game aims to help organisations develop decision-making skills for their workers in early SC stages, so they know the course of action to minimise stockouts when facing DEs in real life. Learning to combine and interpret results from an integral set of key performance indicators (KPIs) is vital to enhance business performance. Patidar et al. (2023) identify 16 KPIs for SC resilience, based on exploration, expert opinion, and an analytical hierarchy process with fuzziness, to prioritise them. Enhancing resilience in the first echelon of an SC would strengthen its foundations and would help to minimise the ripple effect, which describes the impact of a disruption on SC performance and disruption-based scope of changes (Ivanov et al. 2014). Moreover, this work tries to enhance the development of preventive actions to overcome DEs in an SC.

3 Enterprise and supply chain resilience simulation game

Measuring and enhancing people's decision-making skills inside organisations is a robust driver for high efficiency and resilience. This can be achieved through the implementation of the SC simulation game recommended in this work. While playing this game, the player must deal with different DEs during 52 weeks (which is the recommended horizon, but the player has the option to change this setting and adapt it to specific needs), and the participant must make decisions to minimise their adverse consequences and maximise the company's earnings. This simulation game

is inspired by the classic Beer Distribution Game, a role-playing simulation of a four-echelon SC, with the main objective of introducing participants to the notion of system dynamics and computer simulation (Sterman 2000). The SC in the Beer Game contains four sequential elements: a retailer, a wholesaler, a distributor, and a factory, where each company is a client of the preceding company. The retailer is the only one with access to real demand data, since it receives orders from the final customers, and that information is not necessarily shared with the other companies in the SC. Each element of the SC is recommended to be played by a different player. In such a game, changes in the final customer's demand are the only events that occur, and, in the basic level of difficulty, the final customer's demand varies significantly less than the demand (orders) at the other elements of the SC, and this is because of the presence of the bullwhip effect, which occurs when demand variation increases as it goes back in the SC. In the game presented in this work, the player must choose between two suppliers. Still, both of them and the player's company can be affected by multiple events, not only customer's demand fluctuations. It is not intended to model the bullwhip effect since its focus is to model the impact of decisions related to a weekly operation, where the player must make critical decisions, such as selecting the supplier and defining the order quantity, in an environment where internal and external factors influence demand. Those internal factors are commercial ploys that can shift demand significantly, and external factors are meaningful changes in the gross domestic product (GDP). The simulation game also tests decision-making skills related to investments since the player must decide the amount of insurance policies to acquire, depending on their coverages and costs. The proposed game simulates the first echelon of an SC, which represents vital knowledge for practitioners to develop decision-making skills. It is essential to learn SC dynamics step by step, and for practitioners making daily sourcing and inventory decisions, reinforcing this first echelon of their SC would help making their learning process more familiar and smoother. This is the main reason why the proposed game concentrates on the first echelon of an SC. Moreover, focusing in the first echelon of an SC allows for a simple and transparent display of the results and the underlying game logic, which are especially important in intra-organisational gamification applications, as stated in a Delphi study presented by Wunderlich et al. (2020). Expanding in further research the tool's scope to encompass multiple echelons would enhance practitioners with a more holistic understanding of supply chain dynamics and resilience. However, it is important to note that such an enhancement could also increase the complexity of the tool, potentially discouraging its utilisation.

Considering the framework presented in Campos et al. (2024), the proposed simulation game fulfils the recommended steps:

- Define business objective: to make the workforce of a company aware and knowledgeable regarding ESCR, strengthening their decision-making skills.
- Outline target behaviours: enhancing a better decision-making process.
- Describe players: workers from a company, who are working in SC areas, such as Sourcing, Planning, and Inventory Management.
- Determine activity loops: each weekly decision is monitored through KPIs, aiming to motivate the player.

- Entertain players: the game lets players make the same type of decisions as in their real jobs, but since it is a safe environment, engagement and motivation will be created.
- Deploy appropriate tools: in the development of the game, components, and dynamics were carefully chosen, and regarding aesthetics, the final web version (in progress) will consider modern design techniques.

The details of the proposed SC simulation game are as follows: it is based on Comala, a distribution company that imports and commercialises one product, which flows through the SC in cardboard boxes (units). This SC is formed by Comala, its customers, and two suppliers (S1 and S2) offering the same product, thus competing for Comala's purchase orders. Both suppliers have different resilience factors and offer different value propositions, including prices, lead times, and replenishment policies. In Hosseini and al Khaled (2019), the authors define the resilience factors in terms of resilience capabilities: absorbent, adaptative, and restorative. Including resilience factors when choosing partners in an SC is highly recommended.

Every week of the game, the participant must decide the number of boxes to purchase from each supplier, taking into account the company's goals and the suppliers' attributes: prices, inventory policies, and lead times. Moreover, this order quantity definition also considers Comala's internal factors, such as inventory carrying, backorder, and lost sales costs, along with selling price, fill rate, operating expenses, and insurance expenses. The simulation game aims to maximise Earnings before Interests and Taxes (EBIT) for Comala during the playing horizon, where this financial indicator is closely related to the decisions the player must make. The DEs the player can face are high demand variations due to customer behaviour or macroeconomic shifts that affect customers' demand, low fill rate from suppliers, and temporary facilities shutdowns due to natural or human-driven disasters.

The game has been designed using system dynamics deployed in VENSIM software and is in the process of becoming a web application. System dynamics is the simulation technique chosen to develop this game because it can model business dynamics with oscillations, amplification of disturbances, feedback loops, stocks, flows, and nonlinearities created by the interaction of the physical and institutional structure of a system with the decision-making processes of the agents acting within it (Sterman et al. 2015).

Figure 1 shows an example of a supplier's replenishment process, expressed with causal loops. According to (Sterman 2000), causal loop diagrams are an important tool for representing systems' feedback structures. They consist of variables connected by arrows denoting the causal influences among the variables. In Fig. 1, the diagram shows that each supplier can receive orders from Comala's distribution centre. Still, the player must decide the order quantities to purchase from each supplier. Both supplier's inventory replenishment policy consists on a reorder point (calculated as a function of the demand during the lead time and a probabilistic safety stock) and an order-up-to level (QSn—expressed as a function of the lead time demand), which is the maximum quantity of boxes that their factories can deliver to their distribution centres, and thus, is a key driver for the orders they can ship to Comala, along with on hand inventory availability. Like in

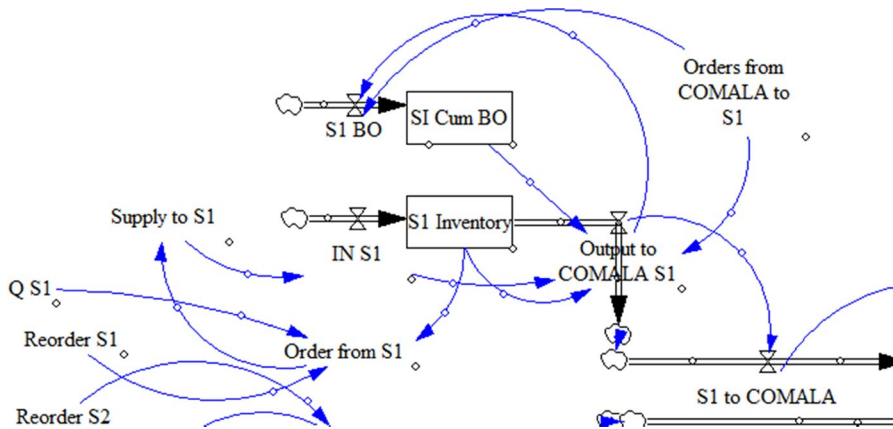


Fig. 1 Inventory replenishment process for a supplier

many real-life situations, both suppliers can accumulate backorders (BO) in their systems if they cannot fulfil the incoming orders on time. When the supplier has sufficient on hand inventory to fulfil BOs, a new shipment to Comala will take place. Specifically, the operating method of the mentioned replenishment model states that when on-hand inventory in the supplier's distribution centre reaches certain level (reorder point), the factory will supply the number of boxes needed to reach the order-up-to level defined in the variable $QS1$. Manufacturing lead times must be considered.

In Fig. 2, there is a diagram with additional variables that affect Comala, from their inventory level available to fulfil customer's demand to their lost sales, back order, and fill rate results.

First, Comala receives shipments from their suppliers, after that, they will try to fulfil their customer's demand. Internal commercial decisions can influence this demand, which is also affected by the economic situation, linked to the GDP; if the increase rate of the GDP is less than 3%, the demand will be severely reduced. The outputs (shipments) from the distribution centre depend on a function that includes the on-hand inventory, the BOs, the customer's demand, and the appearance of a DE (binary variable) that causes the temporal shutdown of the distribution centre.

Table 1 presents the nomenclature and definitions of all the variables involved in the simulation model and, thus, in the game. These variables are classified as level, flow, or auxiliary depending on their function in the model.

The equations explaining the operation of the simulation model are the following:

In all the formulas, the units refer to boxes. Some of the parameters are set to constant natural numbers, they are identified in the equations by X_j .

$$X_j \in \mathbb{N}; \quad j \in \mathbb{N}; \quad j \in [0, 1] \quad (1)$$

Starting from COMALA, the inventory is a level variable named TIC, and its formula is the next one, considering that the initial inventory is set to X_0 units.

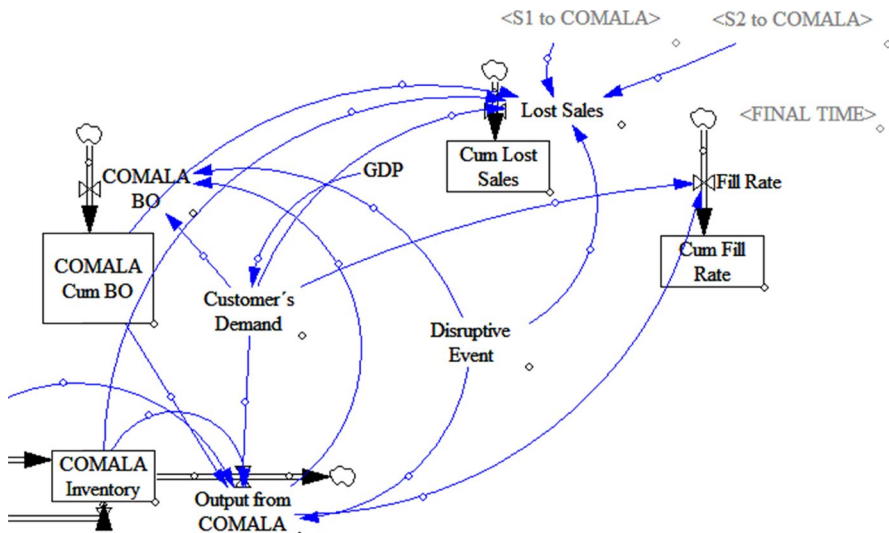


Fig. 2 Causal relations for Comala

$$TIC(t) = \int_0^t (S1C(t) + S2C(t) - OFC(t))dt; \quad TIC(t_0) = X_0 \quad (2)$$

$S1C$ and $S2C$ are "DELAY FIXED" variables that model the delay in the arrival of the units to COMALA during their lead time. The first element in the formula is the number of units delayed, and the second element is the delay duration, measured in weeks ($LS1$ and $LS2$), and for the game, both time variables are set to constant natural numbers.

$$S_iC(t) = DELAY\ FIXED(OTCS_i(t), LS_i(t)); \quad i = 1, 2 \quad (3)$$

Both the suppliers have their warehouses where they keep their inventory. In this model, they are represented as $TIS1$ for supplier 1 and $TIS2$ for supplier 2 and are also level variables. Their equation is the following:

$$TIS_i(t) = \int_0^t (INS_i(t) - OTCS_i(t))dt; \quad TIS_i(t_0) = X_1; \quad i = 1, 2 \quad (4)$$

where INS_i is a "DELAY FIXED" variable modelling the manufacturing lead time (ML) needed to supply a specific supplier from the production plant. Supply to $S1$ and Supply to $S2$ are both set to constant natural numbers.

$$INS_i(t) = DELAY\ FIXED(STS_i(t), MLS_i(t)); \quad i = 1, 2 \quad (5)$$

Table 1 Variables of the simulation model

Nomen- clature	Graph description	Definition
<i>Level variables</i>		
<i>TIC</i>	COMALA Inven- tory	Total inventory at COMALA (Distribution Centre)
<i>TIS_i</i>	<i>S_i</i> Inventory	Total inventory at supplier <i>i</i> (Warehouse)
<i>TBOS_i</i>	<i>S_i</i> Cum BO	Total back orders at supplier <i>i</i>
<i>CCBO</i>	COMALA Cum BO	Total back orders at COMALA
<i>CFR</i>	Cum Fill Rate	Total Fill Rate for COMALA
<i>CLS</i>	Cum Lost Sales	Total lost sales for COMALA
<i>CSOC</i>	Cum stockouts cost	Total inventory stockout cost for COMALA
<i>CIHC</i>	Cum Inventory holding cost	Total inventory holding cost for COMALA
<i>CRC</i>	Cum Replenish- ment cost	Total replenishment cost for COMALA
<i>CIC</i>	Inventory cost	Total inventory cost at COMALA
<i>CREV</i>	Cum revenue	Total revenue for COMALA
<i>CGP</i>	Cum gross profit	Total gross profit for COMALA
<i>CEBIT</i>	Cum EBIT	Total EBIT for COMALA
<i>Flow variables</i>		
<i>S_iC</i>	<i>S_i</i> to COMALA	Delayed output from supplier <i>i</i> to COMALA
<i>OFC</i>	Output from COMALA	Output from COMALA
<i>INS_i</i>	IN <i>S_i</i>	Delayed input to supplier <i>i</i> inventory
<i>OTCS_i</i>	Output to COMALA <i>S_i</i>	Output from supplier <i>i</i> inventory to COMALA
<i>S_iBO</i>	<i>S_i</i> BO	Backorders from <i>S_i</i>
<i>CBO</i>	COMALA BO	Backorders from COMALA
<i>FR</i>	Fill Rate	Unit fill rate from COMALA to their customers
<i>LS</i>	Lost Sales	Weekly lost sales for COMALA
<i>SOC</i>	Stockouts cost	Weekly cost generated by inventory stockouts
<i>IHC</i>	IHC	Inventory holding cost
<i>RC</i>	Replenishment cost	Replenishment cost
<i>UC</i>	Unit cost	Weighted average unit cost
<i>REV</i>	Revenue	Weekly revenue for COMALA
<i>GM</i>	Gross margin	Gross margin for COMALA
<i>GP</i>	Gross Profit	Gross Profit for COMALA
<i>EBIT</i>	EBIT	Earnings before interest and taxes for COMALA
<i>Auxiliary variables</i>		
<i>STS_i</i>	Supply to <i>S_i</i>	Orders supplied from manufacturing to supplier <i>i</i>
<i>OFS_i</i>	Order from <i>S_i</i>	Orders placed from supplier <i>i</i> to manufacturing
<i>LS_i</i>	<i>LS_i</i>	Lead time from supplier <i>i</i> to COMALA

Table 1 (continued)

Nomen- clature	Graph description	Definition
MLS_i	MLS_i	Manufacturing lead time to supply supplier i
D	Customer's demand	Customer's demand
$ROPS_i$	Reorder S_i	Reorder point for supplier i
XSi	Mean X_i	Average number of units per week ordered from Comala to each supplier i
$ssSi$	$ss S_i$	Safety stock for supplier i
$sigSi$	$sigI S_i$	Standard deviation of the number of units per week ordered from Comala to each supplier i
QSI	$Q S_1$	Order-up-to level for supplier 1
$QS2$	$Q S_2$	Order-up-to level for supplier 2
DE	Disruptive Event	Disruptive event
GDP	GDP	Gross Domestic Product (increase rate)
sc	Unit stockout cost	Unit stockout cost
r	Inventory holding cost rate	Inventory holding cost rate
$INSCOV$	Insurance coverage	Nonlinear relations between coverage and premium
$INSC$	Insurances	Insurance cost
$INSPB$	Insurance pay-back	Insurance payback in the presence of a DE
<i>Game variables</i>		
$OFCTS_i$	Orders from COMALA to S_i	Quantity of units ordered from COMALA to supplier i
$INSP$	Insurance premium	Insurance premium as a percentage of unit cost

$STS1$ and $STS2$ are the number of units suppliers 1 and 2 order to the production plant (OFS). The quantities and frequencies of these orders are based on a reorder point ($ROPS$) and an order-up-to inventory policy (QS). Equations (6)–(10) show the logic of this policy.

$$OFS_i(t) = \begin{cases} \text{if } TIS_i(t) \leq ROPS_i(t) : [(QS_i - TIS_i(t))] \\ \text{otherwise} : [0] \end{cases} \quad i = 1, 2 \quad (6)$$

$$ROPS_i(t) = MLS_i * XS_i(t) + ssS_i(t); \quad i = 1, 2 \quad (7)$$

XS_i is the average number of units per week ordered from Comala to each supplier i , and ssS_i refers to the safety stock for each supplier i .

$$ssS_i(t) = 1.28 * \sqrt{MLS_i} * sigS_i(t); \quad i = 1, 2 \quad (8)$$

$sigS_i$ accounts for the standard deviation of the number of units per week ordered from Comala to each supplier i .

$$QS_1(t) = ROPS_1(t) + XS_1(t) \quad (9)$$

$$QS_2(t) = ROPS_2(t) + 2 * XS_2(t) \quad (10)$$

Each supplier can have backorders, which are modelled by flow variables, as shown in the following equation:

$$S_iBO(t) = OFCTS_i(t) - OTCS_i(t); \quad i = 1, 2 \quad (11)$$

$OFCTS_i$ are game variables the player must provide each "week" of the game duration.

These backorders are accumulated for future shipping when the supplier has enough inventory, and this is represented using level variables, as presented in the next formula:

$$TBOS_i(t) = \int_0^t S_iBO(t)dt; \quad TBOS_i(t_0) = 0; \quad i = 1, 2 \quad (12)$$

COMALA can also have backorders, and they are calculated with a flow variable (CBO) and accumulated in a level variable ($CCBO$). The following two formulas show their equations:

$$CBO(t) = \begin{cases} \text{if } DE(t) = [1:D(t)] \\ \text{otherwise: } [D(t) - OFC(t)] \end{cases} \quad (13)$$

$$CCBO(t) = \int_0^t CBO(t)dt; \quad CCBO(t_0) = 0 \quad (14)$$

where DE refers to a binary variable showing a disruptive event's presence. In the next formula, the first element refers to the starting week where the event occurs, and the second element indicates the duration of the event, expressed as the difference between the time when the DE disappears (t_1) minus the time when it started (t_0).

$$DE(t) = PULSE(t_0, \Delta_t); \Delta_t = t_1 - t_0 \quad (15)$$

In the model, D is the weekly demand, and OFC are the number of units shipped from COMALA to their customers. The following two formulas show the relations for each one of them:

$$D(t) = \begin{cases} \text{if } GDP(t) \geq r : [UL] \\ \text{otherwise : } [LL] \end{cases} \quad (16)$$

where GDP refers to the Gross Domestic Product growth rate, which fluctuates from 2.5% to 7.5%, impacting the customer's demand, UL and LL are upper and lower

limits to the weekly demand, and r represents the threshold influencing the demand behaviour.

The output from COMALA (auxiliary variable OFC) represents the number of units shipped from COMALA to their customers, and the following equation models it:

$$OFC(t) = \begin{cases} \text{if } DE(t) = 1 : [0] \\ \text{otherwise} : [MIN(a(t), b(t))] \end{cases} \quad (17)$$

where:

$$a(t) = \sum_{i=1}^2 S_i C(t) + TIC(t) \quad (18)$$

$$b(t) = CCBO(t) + D(t) \quad (19)$$

This simulation game calculates several performance indicators expressed in the model as a combination of level, flow, and auxiliary variables. One of the leading performance indicators used to monitor customer service is the fill rate, and in this game, it is measured as the unit fill rate, which is expressed with the FR variable that continuously calculates the fill rate and the level variable CFR that shows the cumulative fill rate for the simulation horizon.

$$FR(t) = \begin{cases} \text{if } D(t) \geq 0 : [1] \\ \text{otherwise} : \\ \text{if } OFC \geq D(t) : 1 \\ \text{otherwise} : \left[\frac{OFC(t)}{D(t)} \right] \end{cases} \quad (20)$$

$$CFR(t) = \int_0^t FR(t)dt; \quad CFR(t_0) = 0 \quad (21)$$

Another leading performance indicator in supply chains and organisations is the number of lost sales, which in this model is expressed in units through the LS variable and the level variable CLS , which calculates the cumulative value for the horizon.

$$LS(t) = \begin{cases} \text{if } DE(t) = 0 : [0] \\ \text{otherwise} : [MIN(a(t), b(t))] \end{cases} \quad (22)$$

$$CLS(t) = \int_0^t LS(t)dt; \quad CLS(t_0) = 0 \quad (23)$$

The model requires several income and cost calculations related to the SC simulated; these calculations are expressed as combinations of auxiliary, flow, and level variables. The stockout cost is calculated every week and accumulated using the next two equations:

$$SOC(t) = sc * CCBO(t) \quad (24)$$

where sc is a constant (real number) representing the unit stockout cost.

$$CSOC(t) = \int_0^t SOC(t)dt; \quad CSOC(t_0) = 0 \quad (25)$$

Regarding the inventory holding cost, it is calculated weekly by multiplying the inventory holding cost rate (constant r , a real number) by the total inventory at COMALA, and the result is accumulated for 52 weeks in a level variable, as shown in the following equations:

$$IHC(t) = r * TIC(t) * UC(t) \quad (26)$$

$$CIHC(t) = \int_0^t IHC(t)dt; \quad CIHC(t_0) = 0 \quad (27)$$

The flow variable "Replenishment cost (RC)" is the sum of the SOC and the IHC , and it is accumulated in a stock variable. The following two equations express those relations:

$$RC(t) = SOC(t) + IHC(t) \quad (28)$$

$$CRC(t) = \int_0^t RC(t)dt; \quad CRC(t_0) = 0 \quad (29)$$

The inventory unit cost is a weighted average of the stored units at COMALA, considering if they are produced by supplier 1 or supplier 2.

$$UC(t) = \sum_{i=1}^2 UCS_i * S_i C(t) \quad (30)$$

where UCS_i is a constant representing the unit cost for supplier i . The level variable "Inventory cost (CIC)" represents the total inventory cost.

$$CIC(t) = \int_0^t \left(UC(t) - OFC(t) * \frac{UC(t)}{\sum_{i=1}^2 S_i C(t)} \right) dt; \quad CIC(t_0) = 0 \quad (31)$$

On the income side of the profit of COMALA, there is a flow variable named "Revenue" (REV), which is the unit price (constant- UP) times OFC , and this is accumulated for the simulation horizon in a level variable, as shown in the next two equations:

$$REV(t) = UP * OFC(t) \quad (32)$$

$$CREV(t) = \int_0^t REV(t)dt; \quad CREV(t_0) = 0 \quad (33)$$

The following expression presents the gross margin (GM), which is a function of the unit price, the unit cost, and the on hand inventory.

$$GM(t) = \begin{cases} \text{if } UP = 0 : [0] \\ \text{otherwise} : \left[UP - \begin{cases} \text{if } TIC(t) = 0 : 0 \\ \text{otherwise} : \frac{CIC(t)}{TIC(t)} \end{cases} \right] \end{cases} \quad (34)$$

The related gross profit is expressed and accumulated in the following flow and level variables, respectively:

$$GP(t) = UP * GM(t) * OFC(t) \quad (35)$$

$$CGP(t) = \int_0^t GP(t)dt; \quad CGP(t_0) = 0 \quad (36)$$

Insurances are solid options to enhance resilience because of their coverage when DEs occur. The insurance premium ($INSP$) is a game variable; thus, the player can choose the percentage of the premium. The insurance cost is expressed in the next equation:

$$INSC(t) = INSP(t) * OFC(t) * (1 - GM(t)) * UP \quad (37)$$

The insurance coverage ($INSCOV$) has a nonlinear relation with the $INSP$, so it is modelled using a WITH LOOKUP auxiliary variable with the following parameters:

$$INSCOV(t) = \begin{cases} \text{if } INSP(t) = 0 : [0] \\ \text{if } INSP(t) = 0.05 : [0.5] \\ \text{if } INSP(t) = 0.1 : [1] \end{cases} \quad (38)$$

Additionally, the insurance payback is calculated as follows:

$$INSPB(t) = \begin{cases} \text{if } DE(t) = 0 : [0] \\ \text{otherwise} : [INSCOV(t) * UP * GM(t) * LS(t)] \end{cases} \quad (39)$$

The primary indicator of this simulation game is the *EBIT*. It is modelled in a flow variable with that name, and the results are accumulated in a level variable (*CEBIT*), as shown in the following equations:

$$EBIT(t) = GP(t) - OPEX - RC(t) - INSC(t) + INSPB(t) \quad (40)$$

OPEX refers to operating expenses, which is a constant value per unit in this game.

$$CEBIT(t) = \int_0^t EBIT(t)dt; \quad CEBIT(t_0) = 0 \quad (41)$$

Figure 3 shows the relations between the inventory and financial variables that the model utilises to calculate the *EBIT*, defined as earnings before interest and taxes, the game's primary objective.

Each week of the game, the model will calculate Comala's fill rate and all the financial indicators needed to determine the *EBIT*. During the course of the game, the participant can monitor the leading indicators directly from a control panel. The performance indicators plotted in the control panel for Comala's operation are the following:

- Weekly revenue.
- Inventory level.
- Customer's demand.
- Lost sales.
- Orders placed to supplier 1.
- Orders placed to supplier 2.
- Replenishment cost.
- Fill Rate from Comala to customers
- Cumulative revenue.
- Cumulative replenishment cost.
- Cumulative *EBIT*.

The KPIs selected to use in the proposed SC game have direct relation with 7 of the indicators proposed in Patidar et al. (2023), as it is shown in Table 2, ensuring that the KPIs included in the SC game are linked to a formal set of ESCR indicators.

Comala's control panel includes three input sliders (one for each game variable of the systems dynamics model) that the player must employ when making decisions; the first two sliders are for determining the order quantity to purchase from each supplier, and the third slider is for choosing the amount to invest in insurances for disaster coverage. Figures 4 and 5 show examples of the control panel.

Currently, the game is a prototype version; thus, the next step is focused on developing a web application (in progress) to make it easily reachable to people from

companies and universities. After the web version is ready, the next step is to implement it in several real companies to test if employees are properly prepared to make the right decisions considering the variables included in the game. It is important to highlight that the developed gaming tool can be customised to represent several real-life situations, and thus, each company can take advantage of this customisation. Examples of different situations that can be addressed in the proposed gaming tool are the following:

- High customers' demand fluctuations.
- Changes in GDP, in representation of today's macroeconomic variations.
- Impactful DEs' appearances that imply temporary closures of Comala's distribution centre.
- Supply constraints related to capacity and inventory policies for each supplier.
- Investment decisions related to insurances to minimise risks.

This implementation should lead to a robust decision-making process, since the situations the users face in the game are similar to what they face in their organisations. When companies include the proposed game in their formal training programs, the first echelon of their SC would be managed by people with better skills to face challenges regarding inventory management (stockouts), supplier selection, and even, insurance related decisions. Complementing their gaming experiences with formal action plans to deal with DEs, such as stated in Sanchis and Poler (2019), will enhance ESCR.

However, it is worth mentioning that this work has limitations that must be considered when implementing its proposals:

- The logic behind the game is based on a simulation model, and thus, it provides a simplified representation of a complex reality.
- The quantity and variety of DEs included in the game are limited.
- The game is designed for single users, although it can involve a team behind the player making decisions.
- The scope of the game is limited to the initial stage of an SC.
- Only one item is considered in the simulation game flow.
- The company (Comala) has only two suppliers.
- The implementation phase of this work will be carried out in a limited number of organisations, since the game is intended to be deployed in one company at a time, until their results are proven reliable.
- Improvements to the simulation model can be made during the implementation process to ensure a better representation of real-life situation in the business world.

Despite these limitations, the gaming tool proposed in this work represents a value-added tool for organisations, since they can enhance their training and learning processes through a game focused on improving skills to manage the first stage of an SC, which is where everything begins.

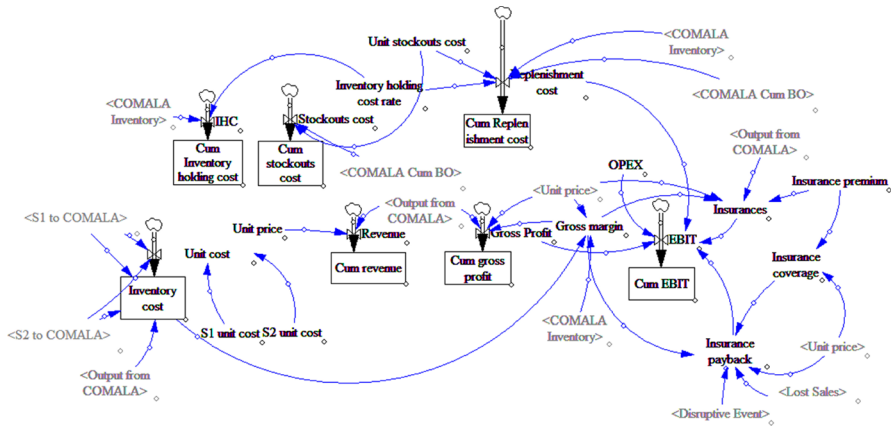


Fig. 3 Cost elements and profit

4 Conclusions

Organisations must endorse resilience as a core component of their value propositions to grow and remain competitive in a complex business world marked by different types of DEs. Training their workforces in the resilience topic will prepare them to make better decisions when facing DEs, and beyond that, evaluating their decision-making skills on this matter will ensure better strategies and ploys to cope with DEs. An ESCR-specialised simulation game, like the one presented in this work, will be helpful for training and evaluation of these decision-making skills since it models SC dynamics and several DEs. This game combines SC scenarios where players have to make tactical decisions to overcome different negative

Table 2 Relations between KPIs

KPIs in the game	Referenced SC KPIs for resilience						
	Lead Time	Supplier delivery efficiency	Demand and supply variations	Service rate	On-time delivery	Inventory velocity	Stock level
Revenue							
Inventory	✓					✓	✓
Demand			✓				
Lost Sales		✓					
Orders to S1							
Orders to S2							
Replenishment cost							
EBIT							
Fill Rate				✓	✓		

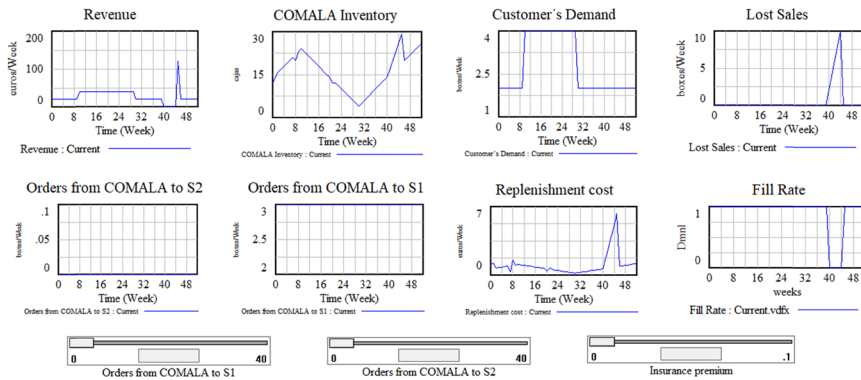


Fig. 4 Control panel, weekly indicators, and sliders

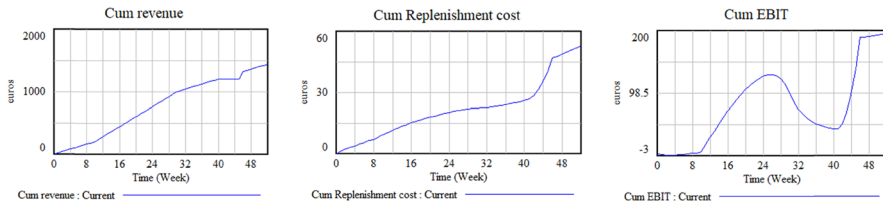


Fig. 5 Control panel, cumulative results

consequences generated by various DEs; this is the main reason why it is appropriate for preparation to face the real world. Another learning topic indirectly addressed in this game is comprehending the concepts and relations among several enterprise key performance indicators, which would strengthen the decision-making process. The main contribution of this work is training people who work in the SC field, through a low cost, low risk methodology which is driven by a simulation game that offers more and different DEs, and thus, different situations compared to the Beer Game. Thus, the main contributions of this research involve: (i) the development of a simulation game designed to train individuals in SC management and resilience principles, providing a low-cost, low-risk environment for skill development; (ii). the identification of important factors influencing resilience in supply chains, such as supplier selection, inventory management, and response strategies to disruptive events; and (iii) the use of key performance indicators (KPIs) in the game for evaluating and improving resilience within supply chain operations.

Other proven training techniques and other types of games can complement the simulation game proposed in this paper, for example, trivia games aiming to enhance awareness of the impact of DE. A complete roadmap to resilience learning can be developed based on the proposed simulation game.

With knowledge and decision-making enhanced through simulation and gamification, the next step for organisations is to develop preventive plans to stay ahead of DEs. Further research will focus on monitoring and predicting DEs, by developing and implementing accessible and easy-to-use tools for organisations to achieve

preparedness capacity. A complete roadmap in the quest for ESCR must be implemented in organisations.

Moreover, further research lines can include prioritising and selecting more common DEs, modelling them in the simulated SC and, thus, having them in the game. Another potential improvement would be adding more key performance indicators to strengthen the decision-making process; these indicators would be advisable if they were selected based on other formal research to determine the most important ones when it comes to enhancing ESCR. After the implementation of the SC game in several companies, their feedback will guide other future developments. These can range from including more echelons to the SC to adding reverse logistics capabilities. Since there are many options, the feedback received from players who work in the field, will be an important driver.

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Data availability The results of this research are available upon request.

Declarations

Conflict of interest The authors declare that there are no financial interests or personal relationships that could have influenced this work.

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